Ball angle in PADD WS

How the direction is chosen at launch and on hits, the TWIST paddle, and a comparison with the 1972 Pong

This document explains how PADD WS decides which way the ball goes: when it is launched from the centre, when a paddle hits it, and with the TWIST paddle of NEXT mode. The logic is the one in the Rust core (`rust/core`, `game.rs` and `next.rs`). The MiSTer and C64 versions reproduce it exactly, tick by tick.

1. Velocity, slope and angle

The ball has a horizontal velocity **vx** and a vertical velocity **vy**, in pixels per tick, with y growing upwards. In this document values refer to the 800×480 field at 60 ticks per second, as in the configuration file. The core rescales them to the actual field and simulation rate: 240 Hz on PC, web and MiSTer, 50 Hz on the C64.

It is easiest to think in terms of the **slope** `s = vy / |vx|`: 0 is a flat shot, ±1 is a 45° shot. Every angle rule works on this ratio.

> The **boost** (EVO, dash, HAMMER, RUSH, SMASH, SUPER) only multiplies horizontal movement: in one tick the ball moves `vx·boost` horizontally and `vy` vertically. A boosted shot is therefore faster and also flatter than its slope suggests.

2. Launch from the centre

Players never serve. After each point the ball waits in the centre for half a second, then starts on its own towards the player who conceded the point. In single-player mode it always goes towards the player. The velocity is random, picked from four combinations:

vx	vy	angle	speed
2	±1	26.6°	2.24
2	±2	45.0°	2.83
3	±1	18.4°	3.16
3	±2	33.7°	3.61

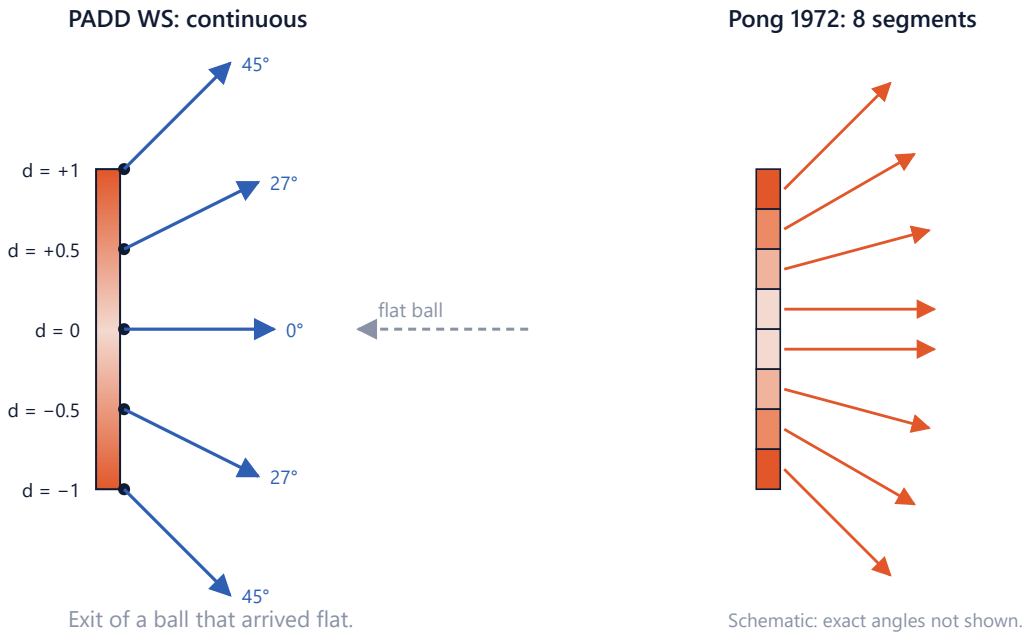
Values in px per tick at 60 Hz. The sign of vy (up or down) is random.

3. Paddle hits

On a hit, what matters is where the ball touches the paddle. Call **d** the distance of the contact point from the paddle centre, divided by half the paddle height: 0 at the centre, +1 at the top edge, −1 at the bottom edge (slightly beyond when the ball catches the corner). The core does, in order:

```
1. vy ← vy + |vx| · d          (that is, s ← s + d)
2. if |vy| > |vx|: vy ← ±|vx|   (never steeper than 45°)
3. vx ← −vx
4. if |vx| is below the maximum speed: vx ← vx·k, vy ← vy·k
```

Step 4 speeds the ball up without changing its angle, because vx and vy are multiplied by the same k. At normal difficulty k is 1.25 in two-player mode (maximum speed 8), 1.10 in single-player and NEXT (maximum 10).



What this means in play

- **The contact point adds to the incoming angle.** Hit at the centre (d = 0), the ball goes back with the same slope it arrived with, like a mirror.
- **The edges change the slope by ±1.** A flat ball caught at the edge leaves at 45°; caught halfway between centre and edge it leaves at about 27°.
- **To flatten a steep ball** you hit it on the opposite side: a ball rising at 45° (s = +1) hit on the bottom edge (d = −1) leaves flat.
- **Speed does not matter.** The deflection is proportional to |vx|, so the same spot on the paddle gives the same change in slope at any speed.
- **Paddle movement does not matter**, except with TWIST (section 4).

Incoming ball	d = −1	d = −0.5	d = 0	d = +0.5	d = +1
flat (s = 0)	45° down	27° down	0°	27° up	45° up
27° up (s = +0.5)	27° down	0°	27° up	45° up	45° up
45° up (s = +1)	0°	27° up	45° up	45° up	45° up

Special cases

- **Single-player wall.** The wall on the right is a paddle as tall as the field: d is measured from the centre of the field. The wall returns the ball without speeding it up.
- **EVO boost area.** In the grey band at the centre of the paddle (a quarter of its height at first, narrower after each boost, down to a twentieth) d is measured from the band centre against half the band, and counts one fifth: the slope changes by ±0.2 at most (about 11°). In exchange the ball leaves with a ×2 boost, which makes it even flatter.
- **Dash interception.** It ignores the contact point: vx grows by a fifth and vy gets a small random deflection (up to ±1 px per tick).

4. The TWIST paddle (NEXT)

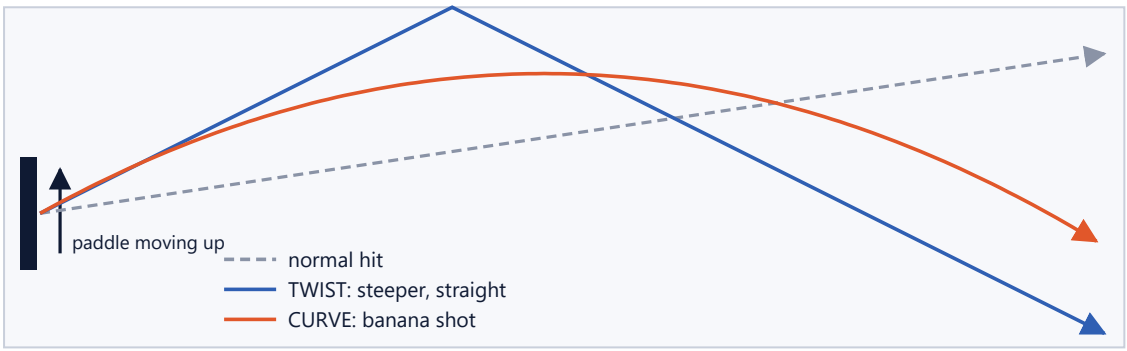
TWIST is the only paddle type where **movement** matters too. For every paddle the core keeps the vertical movement per tick, smoothed:

```
motion ← motion + (Δy − motion) / 4    (every tick, encoder included)
```

After the normal hit of section 3, with vx already reversed and sped up, TWIST adds half of this movement:

```
vy ← vy + motion · 0.5,  limited to ±|vx|   (always within 45°)
```

- **Direction.** The ball follows the paddle: hitting while moving up sends it higher, hitting while moving down sends it lower. This adds to the contact-point effect.
- **Strength.** At full speed the paddle adds 7.5 px per tick (at 60 Hz). While vx is below 7.5, movement alone is enough to turn the shot to 45°. With a faster ball it matters less.
- **Timing.** Movement is smoothed: stopping just before the hit cancels almost all of the effect (after 1/30 of a second about 10% is left). You have to hit while moving.
- **Straight path.** TWIST only changes the exit angle: after that the ball flies straight.
- **Priority.** SMASH, SUPER and the special moves (STRAIGHT SHOT, RISING STRIKE) choose the angle after TWIST, so they override it.



The same hit with the normal paddle, with a moving TWIST and with the CURVE bonus.

Other NEXT shots that set the angle

Shot	Angle	In flight
CURVE (bonus)	that of the hit	curves against its vertical direction (0.3 px/tick², at 60 Hz); the curve flips when the ball bounces off an edge; always within 45°
STEER (bonus)	that of the hit	the hitter steers it with up and down until 3/4 of the field, up to about 63°; their paddle stays still
SMASH (bonus)	slope 3/4, towards the side away from the opponent's paddle	straight, ×2.2 boost
SUPER (bonus)	45°, towards the side away from the opponent's paddle	straight, ×3.2 boost
STRAIGHT SHOT (move)	flat	straight, ×1.5 or ×2 boost
RISING STRIKE (move)	flat	at 5/8 or 7/8 of the field it turns to 45° towards the edge away from the opponent's paddle

5. Comparison with the original Pong

The 1972 arcade Pong was built from discrete logic, with no processor. Its rules are given here as the sources on its hardware describe them.

	Pong (1972)	PADD WS
Contact point	paddle split into 8 segments, each with a fixed exit angle	continuous: d from −1 to +1, in Q16.16 fixed point
Incoming angle	ignored: only the segment decides	kept: the deflection adds to the incoming slope
Centre hit	the ball leaves flat	the ball returns with its incoming angle
Edge hit	outer segments give the widest angles	slope ±1 added
Maximum angle	that of the outer segments	45° (about 63° only with STEER)
Paddle movement	ignored	ignored; with TWIST half of it goes to the ball
Spin in flight	none	CURVE and STEER in NEXT
Ball speed	grows during a rally, resets after a point	×k on every hit up to a maximum, back to 2–3 px/tick after a point
Launch	automatic	automatic, towards the player who conceded, with 4 random angles
Top edge	paddles cannot reach the top, so long rallies end	paddles cover the whole field

Bending the ball in flight does not come from Pong but from the Magnavox Odyssey (1972): its controllers had a knob called *english*, the billiards word for spin, which changed the ball's vertical path while it was in the air.

Source for the original Pong: the Wikipedia article “Pong” (en.wikipedia.org/wiki/Pong), which reports the descriptions of the Atari hardware. The PADD WS rules are those of the Rust core in version 4.0.0-beta.16.